STUDENT PROJECT BRIEF

Contactless Vital-Sign Detection from Thermal Video to Support Emergency-Department Handover

Image Processing and Handling

Primary focus	Contactless estimation of respiratory rate from radiometric thermal video
Setting	Simulated emergency-department handover or admission
Project type	Semester-long, offline proof of concept
Core result	Reproducible processing pipeline and quantitative comparison with a synchronized reference

Background and Project Objective

Patient handovers and admission procedures in emergency medicine must transfer clinically relevant information quickly while diagnostic and monitoring activities continue in parallel. Vital signs are a central part of this information, but their acquisition still requires time, attention, and physical interaction with the patient. The broader scientific vision is to support faster and more structured handovers by automatically acquiring selected observable patient parameters and making them available as structured information during the transition of care. Such systems should support, rather than replace, clinical assessment and communication.

This vision is particularly relevant at the interface between emergency medical services and the emergency department. Handover is vulnerable to interruptions, information loss, differences in documentation, and competing clinical tasks [1]. Contactless acquisition could reduce the need to perform every measurement sequentially and could allow selected parameters to be prepared while the verbal handover is taking place. However, before such an approach can be considered for clinical use, its technical feasibility and limitations must be investigated under controlled conditions.

Respiratory rate is a suitable starting point for this methodological development. It is clinically important but is commonly determined by manual counting or indirectly derived from monitoring systems. Thermal cameras can capture temperature changes around the nose and mouth caused by alternating inhaled and exhaled air, as well as respiratory movements of the upper body. A recent review structures thermal respiration monitoring into data acquisition, definition and tracking of an anatomical region of interest, extraction of a respiratory signal, and estimation of respiratory parameters [2].

Initial clinical studies support the feasibility of this approach. Aldred et al. estimated respiratory rate from temperature fluctuations around the nostrils in 136 patients during emergency-department admission and reported good agreement with manual chest-movement counting [3]. Chan et al. evaluated infrared-video-based respiratory-rate estimation in a critically ill inpatient population [4]. Algorithmic work has further demonstrated that the nasal region can be tracked and respiratory signals extracted even when thermal contrast and head position vary [5]. These studies also show why automatic target selection,

robust region-of-interest tracking, motion handling, and explicit signal-quality assessment are essential.

Thermal imaging may provide further physiological information, but the maturity of the individual measurements differs. Cardiorespiratory signals have been explored in healthy participants using thermal imaging [6], and combined thermal and RGB systems have been used to estimate respiratory rate, heart rate, and temperature-related parameters [7]. Heart-rate estimation is more demanding because the relevant thermal signal is weak and sensitive to motion. Likewise, a thermal camera measures skin-surface radiation; it does not directly measure core body temperature. These parameters should therefore be treated as optional extensions that require separate references and careful validation.

A simulation laboratory provides an appropriate environment for the first development stage. Camera position, recording distance, room temperature, participant roles, movement, and reference measurements can be defined prospectively, and recordings can be repeated if technical problems occur. Simulation-based research at the institute has already demonstrated how audiovisual observations can be transformed into structured, computer-generated information for feedback [8]. The present project applies the same general principle of automated observation to physiological measurement during a simulated handover or admission, rather than to the assessment of learner behaviour.

The project will therefore develop and evaluate an offline image-processing workflow that identifies the principal patient in thermal video, tracks an appropriate facial or upper-body region, and estimates respiratory rate from the resulting time series. The students will define and justify their own technical approach and project plan. They will also design a realistic simulated handover or admission scenario, specify the recording and reference-measurement procedure, and collect the required data in the institute's simulation laboratory.

The work is a semester-long proof-of-concept project, not the development of a certified medical device or a clinically deployable monitoring system. It should use simulated scenarios and consenting volunteers, not real patients. The primary contribution is a reproducible pipeline and an empirical assessment of whether respiratory rate can be recovered under the defined conditions. The project should also document when the method fails and which requirements would have to be addressed before later clinical evaluation.

Research Questions

Main Research Question

> **Can a thermal-video-based workflow automatically identify and track the principal patient and estimate respiratory rate with sufficient technical accuracy and availability in a controlled, simulated emergency-department handover or admission?**

Secondary Research Questions

1. **Patient Identification and Tracking.** How reliably can the principal patient be selected and tracked when other people are present in the scene?
2. **Respiratory-Signal Extraction.** Which anatomical region and signal-processing approach provide a stable respiratory signal under the selected recording conditions?
3. **Accuracy and Robustness.** How closely does the estimated respiratory rate agree with a synchronized reference, and how is performance affected by movement, speaking, distance, or head orientation?
4. **Extension Potential.** Which additional measurements, such as heart rate or facial skin temperature, appear technically feasible for subsequent work?

Outcomes

Primary Outcome

> Agreement between thermal-video-derived respiratory rate and a synchronized reference measurement. The evaluation should report suitable error and agreement measures, such as mean absolute error, bias and limits of agreement, together with the proportion of recording windows for which the algorithm produces a valid estimate.

Secondary Outcomes

- Successful identification and temporal tracking of the intended patient, including transparent criteria for defining the principal person in the scene.
- Technical performance under prespecified scenario variations and a structured description of failure cases and missing estimates.
- A reproducible dataset description, executable processing pipeline, and documented recording protocol that can support later development.

Minimum Result

Minimum successful result. An offline pipeline that processes radiometric thermal recordings, automatically identifies and tracks at least one principal participant over a defined sequence, extracts a respiratory signal, estimates respiratory rate for predefined measurement windows, and compares the estimates quantitatively with a synchronized reference. The code and recording protocol must be documented and reproducible.

Tasks

01 Literature Review and Scope Definition

- Review relevant work on contactless vital-sign acquisition, thermal respiration monitoring, person and facial-region tracking, physiological signal extraction, and evaluation of measurement agreement.
- Define the precise technical scope, success criteria, minimum result, and optional extensions for the semester project.

02 Algorithmic Concept and Project Plan

- Develop and justify an algorithmic approach for detecting people, selecting the principal patient, tracking the relevant region of interest, extracting a respiratory signal, and estimating respiratory rate.
- Translate the proposed approach into an independent work plan with work packages, responsibilities, milestones, risks, and fallback options.

03 Simulation and Data-Acquisition Design

- Design a realistic camera and recording setup for a simulated emergency-department handover or admission, including the thermal camera position, relevant field of view, measurement distance, and reference-signal acquisition.
- Define a standardized script covering the room setting, participant roles, sequence of events, patient position, movements, speaking phases, test conditions, and data-collection procedure.
- Specify how thermal data, timing information, and reference measurements will be synchronized, stored, and documented.

04 Data Acquisition in the Simulation Laboratory

- Set up and test the planned recording procedure in the institute's simulation laboratory and revise it if pilot recordings reveal technical limitations.
- Conduct the planned data acquisition with consenting volunteers, including calibration or development recordings and separate recordings reserved for evaluation.

05 Implementation and Evaluation

- Implement the proposed workflow using a justified technical stack and make the processing steps reproducible from the original recordings.
- Evaluate patient selection, tracking availability, respiratory-rate accuracy, and relevant failure cases using the predefined reference data and test conditions.

Optional extension. If the minimum result is achieved, investigate one clearly defined extension, such as motion robustness, multi-person scenes, heart-rate estimation, facial skin-temperature analysis, RGB-thermal fusion, or near-real-time processing.

06 Synthesis and Reporting

- Summarize the technical feasibility of automated respiratory-rate acquisition in the simulated handover or admission setting and distinguish demonstrated findings from future clinical claims.
- Discuss limitations related to the controlled setting, participant sample, camera characteristics, reference measurements, environmental conditions, and transferability to real emergency-department workflows.
- Prepare the final scientific presentation and a documented archive containing executable code, environment information, data and recording documentation, evaluation outputs, and a concise implementation outlook.

References

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[2] Pires Alves R, van Meulen F, Overeem S, Zinger S, Stuijk S. Thermal Cameras for Continuous and Contactless Respiration Monitoring. *Sensors*. 2024;24(24):8118. doi:10.3390/s24248118.

[3] Aldred A, Ribeiro JAS, Bezerra PMS, Antunes ACM, Goulart AC, Desuo IC, Gomes G. Application of thermography to estimate respiratory rate in the emergency room: The journal Temperature toolbox. *Temperature*. 2023;10:159–165. doi:10.1080/23328940.2022.2099215.

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